Complex Analysis
Sections 7.1–7.4

MA4410

Pointwise Convergence

Let $\{f_n\}$ be a sequence of functions on a set E.

We say $f_n \to f$ **pointwise on** E if

$$\lim_{n\to\infty} f_n(z) = f(z) \quad \text{for each } z \in E.$$

Convergence occurs separately at each point.

Uniform Convergence

We say $f_n \to f$ **uniformly on** E if

$$\forall \varepsilon > 0 \; \exists N \text{ such that } |f_n(z) - f(z)| < \varepsilon$$

for all $n \geq N$ and **all** $z \in E$.

The same N works for every point in E.

Absolute Convergence of a Series

A series of functions

$$\sum_{n=1}^{\infty} f_n(z)$$

converges **absolutely** at z if

$$\sum_{n=1}^{\infty} |f_n(z)|$$

converges.

Absolute convergence $\Rightarrow$ convergence.

Why Uniform Convergence Matters

Pointwise convergence alone is not strong enough to guarantee:

- Continuity of the limit
- Interchange of limit and integral
- Interchange of limit and derivative

Uniform convergence allows limits to pass through integrals and (with additional hypotheses) derivatives.

Uniform Convergence and Differentiation

Key Principle:

If $\{f_n\}$ are analytic on a region and

- $f_n \to f$ uniformly on compact subsets, and
- f_n' converges uniformly on compact subsets,

(In this context a set is compact iff it is closed and bounded.)
then

$$f'(z) = \lim_{n\to\infty} f_n'(z).$$

Thus differentiation may be performed term-by-term.

Taylor Series

A Taylor series centered at z_0 is

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n.$$

Here

$$a_n = \frac{f^{(n)}(z_0)}{n!}.$$

- Contains only nonnegative powers
- Converges in a disk
- Represents an analytic function inside its radius of convergence

Laurent Series

A Laurent series centered at z_0 is

$$\sum_{n=-\infty}^{\infty} a_n(z - z_0)^n.$$

It can be written as

$$\sum_{n=0}^{\infty} a_n(z - z_0)^n + \sum_{n=1}^{\infty} \frac{a_{-n}}{(z - z_0)^n}.$$

Convergence of a Laurent Series

The Laurent series converges at z if

$$\sum_{n=-N}^{N} a_n(z-z_0)^n$$

has a limit as $N \to \infty$.

Region of Convergence

There exist numbers R_1, R_2 with

$$0 \leq R_1 < R_2 \leq \infty$$

such that the series converges absolutely in

$$R_1 < |z - z_0| < R_2$$

and diverges outside.

Uniform Convergence for Laurent Series (7.1.1)

If

$$R_1 < |z - z_0| < R_2,$$

then the Laurent series converges uniformly on every closed annulus

$$r_1 \leq |z - z_0| \leq r_2$$

with

$$R_1 < r_1 < r_2 < R_2.$$

Hence the sum is analytic in the annulus.

Term-by-Term Differentiation

Inside the annulus,

$$\frac{d}{dz}\left(\sum_{n=-\infty}^{\infty} a_n(z-z_0)^n\right) = \sum_{n=-\infty}^{\infty} na_n(z-z_0)^{n-1}.$$

The derivative has the same region of convergence.

Application to Laurent Series

A Laurent series converges uniformly on closed annuli inside its region of convergence.

Therefore:

- The sum is analytic.
- It may be differentiated term-by-term.
- The derivative has the same region of convergence.

Uniform convergence is the engine that makes complex power and Laurent series so powerful.

Isolated Singularities

z_0 is an isolated singularity if

$$f \text{ is analytic on } 0 < |z - z_0| < R.$$

Classification via Principal Part

From the Laurent expansion:

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k.$$

- All $a_{-k} = 0$ for $k > 0 \Rightarrow$ removable singularity
- If $a_{-n} \neq 0$ and $a_{-k} = 0$ for $k > n$ then

  $$f(z) = \sum_{k=-n}^{\infty} a_k (z - z_0)^k.$$

  This is a nth order pole and the principal part is $a_{-n} z^{-n}$.
- The function has an essential singularity if there are always non-zero terms $a_{-k} z^{-k}$ further out in the negative direction. In other words, you can not truncate the sum!

Residue

$$\mathrm{Res}_{z=z_0} f(z) = a_{-1}.$$

Residue Theorem Preview

Residues allow computation of contour integrals:

$$\int_C f(z)\,dz = 2\pi i \sum \text{Res}.$$

Residue Theorem

If f is analytic inside and on C except at isolated singularities z_k, then

$$\int_C f(z)\,dz = 2\pi i \sum_k \mathrm{Res}_{z=z_k} f(z).$$

Big Picture

- General convergence definitions
- Laurent convergence on annuli
- Singularities classified by principal part
- Residue Theorem reduces integrals to algebra

Taylor Series: Definition

Definition (Taylor Series)
Let f be analytic in a disk

$$D(a, R) = \{z : |z - a| < R\}.$$

Then f has a Taylor series expansion about a:

$$f(z) = \sum_{n=0}^{\infty} a_n (z - a)^n.$$

Coefficient Formula

$$a_n = \frac{f^{(n)}(a)}{n!}.$$

The series converges for $|z - a| < R$.

Existence of Taylor Series

Theorem (Existence of Taylor Series)
If f is analytic in $D(a, R)$, then

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(z-a)^n.$$

Key Point
The coefficients are determined by derivatives at a.

Strong Local Control
Knowledge of all derivatives at a single point determines the function in the entire disk.

Laurent Series: Definition

Definition (Laurent Series)
Let f be analytic in an annulus

$$A = \{z : r < |z - a| < R\}.$$

Then a Laurent series about a is

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n.$$

We decompose:

$$f(z) = \underbrace{\sum_{n=0}^{\infty} c_n (z-a)^n}_{\text{analytic part}} + \underbrace{\sum_{n=1}^{\infty} \frac{c_{-n}}{(z-a)^n}}_{\text{principal part}}.$$

This expansion is valid for $r < |z - a| < R$.

Existence of the Laurent Series

Theorem (Existence of Laurent Series)
If f is analytic in
$$A = \{z : r < |z - a| < R\},$$
then f admits a Laurent expansion
$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n$$
that converges for $r < |z - a| < R$.

Important Distinction
Unlike Taylor series, we do *not* yet have a formula for c_n.

Proof of Existence (Outline)

Let γ_1 be a circle of radius ρ_1 and γ_2 a circle of radius ρ_2 with

$$r < \rho_1 < |z-a| < \rho_2 < R.$$

By Cauchy's Integral Formula applied twice,

$$f(z) = \frac{1}{2\pi i}\int_{\gamma_2}\frac{f(\zeta)}{\zeta - z}d\zeta - \frac{1}{2\pi i}\int_{\gamma_1}\frac{f(\zeta)}{\zeta - z}d\zeta.$$

Expand:

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - a}\cdot\frac{1}{1-\frac{z-a}{\zeta-a}}$$

as a geometric series in two different regions.

This produces positive and negative powers of $(z-a)$.

Why Does Cauchy Give This Representation of $f(z)$?

Fix z with

$$r < |z - a| < R.$$

Choose radii ρ_1, ρ_2 so that

$$r < \rho_1 < |z - a| < \rho_2 < R.$$

Consider the region between the two circles:

$$\Omega = \{\zeta : \rho_1 < |\zeta - a| < \rho_2\}.$$

Key Observation

The function

$$\zeta \mapsto \frac{f(\zeta)}{\zeta - z}$$

is analytic on Ω:

- f is analytic in the annulus,
- $\zeta - z \neq 0$ on Ω because z lies strictly between the circles.

Therefore, by Cauchy's Theorem applied to the boundary of Ω:

Extracting $f(z)$ from Cauchy

Now enlarge γ_2 slightly so that it encloses z.

Then by the Cauchy Integral Formula:

$$\frac{1}{2\pi i}\int_{\gamma_2}\frac{f(\zeta)}{\zeta - z}d\zeta = f(z).$$

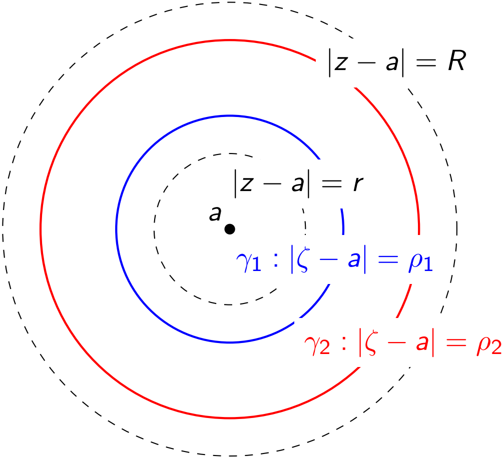
But γ_1 does *not* enclose z so....

$$\frac{1}{2\pi i}\int_{\gamma_1}\frac{f(\zeta)}{\zeta - z}d\zeta = 0.$$

$$f(z) = \frac{1}{2\pi i}\int_{\gamma_2}\frac{f(\zeta)}{\zeta - z}d\zeta - \frac{1}{2\pi i}\int_{\gamma_1}\frac{f(\zeta)}{\zeta - z}d\zeta.$$

Underlying fundamental decomposition for Laurent series.

Geometry of the Laurent Series Proof



$$r < \rho_1 < |z-a| < \rho_2 < R$$

We decompose $f(z)$ using Cauchy integrals over γ_2 and γ_1.

Geometric Series in the Two Regions

From Cauchy's formula:

$$f(z) = \frac{1}{2\pi i}\int_{\gamma_2} \frac{f(\zeta)}{\zeta - z}\,d\zeta - \frac{1}{2\pi i}\int_{\gamma_1} \frac{f(\zeta)}{\zeta - z}\,d\zeta.$$

Outer Contour γ_2 (Positive Powers)
If $|\zeta - a| = \rho_2 > |z - a|$, then

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - a}\,\frac{1}{1 - \frac{z-a}{\zeta-a}} = \frac{1}{\zeta - a}\sum_{n=0}^{\infty}\left(\frac{z-a}{\zeta-a}\right)^n.$$

Valid because

$$\left|\frac{z-a}{\zeta-a}\right| < 1.$$

This produces the **nonnegative powers**.

From Cauchy's formula:

$$f(z) = \frac{1}{2\pi i}\int_{\gamma_2} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i}\int_{\gamma_1} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Inner Contour γ_1 (Negative Powers)
If $|\zeta - a| = \rho_1 < |z - a|$, then

$$\frac{1}{\zeta - z} = -\frac{1}{z-a}\frac{1}{1 - \frac{\zeta - a}{z - a}} = -\frac{1}{z-a}\sum_{n=0}^{\infty}\left(\frac{\zeta - a}{z - a}\right)^n.$$

Valid because

$$\left|\frac{\zeta - a}{z - a}\right| < 1.$$

This produces the **negative powers**.

What We Do NOT Yet Have

For Taylor series:

$$a_n = \frac{f^{(n)}(a)}{n!}.$$

Derivatives at one point determine all coefficients.

For Laurent series:

$$f(z) = \sum_{n=-\infty}^{\infty} c_n(z-a)^n.$$

We know the expansion exists.

But how do we compute c_n?

There is no derivative formula when $n < 0$.

Cauchy Integral Formula for Laurent Coefficients

Theorem (Coefficient Formula)
If f is analytic in $r < |z - a| < R$, then

$$c_n = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - a)^{n+1}} \, d\zeta,$$

where γ is any positively oriented circle in the annulus.

Remark
When $n \geq 0$:

$$c_n = \frac{f^{(n)}(a)}{n!}.$$

When $n < 0$:

$$c_{-1} = \frac{1}{2\pi i} \int_\gamma f(\zeta) \, d\zeta.$$

This is the bridge to residues.

Conceptual Transition

Taylor theory:

Derivatives determine coefficients.

Laurent theory:

Integrals determine coefficients.

Philosophical Shift

Local differential data $\to$ Global contour data

This shift is the foundation of:

- Residue theory
- Classification of singularities
- The Residue Theorem